

Topology as a Computational Basis for ICL in First-Order CRNs

A concise account of the theory, experiments, audit, and current evidence

Topology worktree: `topology`

Evidence source commit: `60de635`

Markov exact-control update: `7067f50`

Multi-base analysis update: `60de635`

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Summary. For first-order CRNs with exponential input-dependent rates, topology is not just a container for trainable parameters. It determines the rooted-spanning-tree sums through which edge projections enter the steady state. The computational basis is therefore not the set of edge vectors $\{K_e\}$, but the set of tree-sum vectors

$$\Theta_T = \sum_{e \in T} K_e, \quad T \in \mathcal{T}_r(G), \quad r \in V.$$

The experiments support the scoped statement that, in the tested fixed-count regimes, topology-associated structural and functional variables explain residual novel-class ICL variation that raw trainable count does not explain. The exact-control phase sharpens this: repaired no-bias γ_{ICL}^* passes analytic toy gates but is not yet a standalone predictor, prospective same-root tree-difference overlap did not survive the first exact causal control, and the completed multi-base exact-control analysis now identifies rooted-tree abundance plus base controls as the clearest current pre-training signal. The strongest evidence remains post-training mechanism evidence: statistic-preserving branch/projection/tree scrambles collapse selected trained models. There is still no universal scalar topology law.

Scope. Everything below is about first-order CRNs with exponential rate encoding. The matrix-tree theorem makes this case exact. Autocatalytic and WTA CRNs need separate topology theories.

[1] The starting question

The original CRN-ICL paper showed that a chemical network can do in-context learning without transformer attention. The input is a context/query vector

$$z = (z_1, \dots, z_{N_c}, z_q), \quad z_i \in \mathbb{R}^D,$$

and the target is the context position whose item matches the query. The true metric is novel-class ICL accuracy, not training accuracy or ordinary validation accuracy.

The paper’s mechanism was subspace projection. For each context position i , define a comparison region

$$\mathcal{M}_i = \{z : z_i \sim z_q\}.$$

ICL succeeds when inputs from different comparison branches produce steady states that the decoder separates.

The original degree-count story was broadly right:

$$n_{\text{req}} = 2N_c(N_c + 1)D, \quad n_g = N_n(N_n - 1)(N_c + 1)D$$

for fully connected first-order CRNs. But this count does not say where the degrees of freedom sit. The topology question was therefore

At fixed count, does the reaction/input topology change ICL?

[2] The exact first-order object

Let $G = (V, E)$ be a strongly connected directed first-order reaction graph. For edge $e : u \rightarrow v$, use exponential rate encoding

$$k_e(z) = \exp(b_e + K_e^\top z).$$

The first-order steady state is controlled exactly by the matrix-tree theorem:

$$\bar{C}_r(z) = \frac{\tau_r(z)}{\sum_s \tau_s(z)}, \quad \tau_r(z) = \sum_{T \in \mathcal{T}_r(G)} \prod_{e \in T} k_e(z).$$

Substitute the exponential form:

$$\prod_{e \in T} k_e(z) = \exp \left(\sum_{e \in T} b_e + \left(\sum_{e \in T} K_e \right)^\top z \right).$$

Set

$$\beta_T = \sum_{e \in T} b_e, \quad \Theta_T = \sum_{e \in T} K_e.$$

Then

$$\tau_r(z) = \sum_{T \in \mathcal{T}_r(G)} \exp(\beta_T + \Theta_T^\top z).$$

This is the key point. A first-order CRN does not read out isolated edge projections $K_e^\top z$. It reads out log-sum-exp combinations of rooted-tree projections $\Theta_T^\top z$.

[3] What topology changes

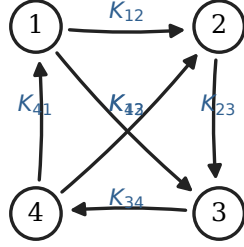
Stack tree incidences as row vectors $s_T \in \{0, 1\}^m$. Then

$$\Theta_T = K^\top s_T.$$

Since concentrations are normalized, common shifts of all tree scores cancel. The more relevant structural object is the tree-difference matrix

$$D_G = \begin{bmatrix} s_{T_1} - s_{T_0} \\ s_{T_2} - s_{T_0} \\ \vdots \end{bmatrix}.$$

physical graph $G = (V, E)$



matrix-tree

computational basis

$$au_r(z) = \sum_{T \in \text{Trees}_r(G)} e^{\beta_T + \Theta_T^\top z}$$

$$\Theta_T = \sum_{e \in T} K_e$$

basis is $\{\Theta_T : T \in \text{Trees}_r(G), r \in V\}$

not the isolated edge set $\{K_e : e \in E\}$

edges carry learned vectors K_e

Figure 1: Topology changes the projection basis. Edge vectors K_e are learned locally, but the steady state uses rooted tree-sum projections Θ_T .

A first count of relative projection capacity is

$$d_{\text{rel}}(G) = p \text{rank}(D_G), \quad p = (N_c + 1)D.$$

With an input mask $\Omega_{e\alpha}$, the capacity must be input-aware:

$$d_{\text{rel}}(G, \Omega) = \sum_{\alpha=1}^p \text{rank}(D_G \text{diag}(\Omega_{\cdot\alpha})).$$

But this is still only a rank proxy. ICL asks a sharper question:

Can the allowed tree-sum directions separate the comparison branches?

The capacity object should be closer to

$$\gamma^*(G, \Omega) = \max_{K, B} \min_{z \in \mathcal{B}} \left[q_{\ell^*(z)}(z) - \max_{\ell \neq \ell^*(z)} q_{\ell}(z) \right]$$

under norm constraints. In the tropical limit,

$$\log \tau_r(z) \approx \max_{T \in \mathcal{T}_r(G)} (\beta_T + \Theta_T^\top z),$$

so the theory becomes a rooted-tree-polytope normal-fan coverage problem.

The hierarchy that emerged is therefore

raw count < relative tree rank < masked tree geometry

< branch-margin/tree-polytope geometry < trained organization < causal dependence.

[4] What was built and audited

The worktree now contains code and reports for arbitrary first-order directed topologies, input masks, tree metrics, branch-capacity probes, post-training tree diagnostics, ablations, motif extraction, and statistic-preserving causal scrambles.

The audit report `ICL/results/next_phase_stats/topology_goal_completion_audit.md` maps the goal to concrete artifacts. It records that the verifier passed for committed artifacts, that hard regimes have full result/model/mechanism/causal follow-ups, and that all headline claims are scoped to `test_novel_classes`. The audit also marks the branch-margin theory as partial by design: proxies exist, but there is not yet a final theorem.

The main source reports used here are:

<code>topology_icl_complete_explanation.md</code>	full narrative explanation
<code>next_phase_evidence_report.md/json</code>	fixed-count, motif, causal summaries
<code>mechanism_isolation_evidence.csv</code>	controlled mechanism contrasts
<code>stat_preserving_causal_stratified</code>	stratified causal scrambles
<code>degree_rewire_normal_fan...</code>	exact-degree normal-fan pilot

[5] Fixed-count structural evidence

The original controlled regime fixed

$$N_n = 6, \quad m = 20, \quad \text{input-coupled count} = 200.$$

It used three physical backbones, 16 selected masks/topologies per backbone, five training seeds each, 240 total runs, and 48 topology/mask groups.

At group level, with target equal to mean novel-class ICL accuracy, the leave-one-out values were

model	group LOO R^2	interpretation
raw count	-0.043	no explanatory power at fixed count
raw count + d_{rel}	0.145	relative rank helps
masked tree geometry	0.189	input-aware geometry helps
tree geometry	0.409	strongest structural summary here

The corresponding hard $N_n = 5, m = 12, N_c = 3, D = 2$ regime showed a different but still positive pattern: raw count and d_{rel} alone were both -0.190, tree geometry was 0.324, and masked tree geometry was 0.447.

The safe conclusion is not that d_{rel} is the law. It is that topology-associated variables explain residual ICL variation that raw count cannot explain in the tested regimes.

[6] Why input masks matter

Physical topology and input-encoding topology are distinct. The physical graph fixes the rooted trees:

$$G_{\text{phys}} \mapsto \mathcal{T}_r(G).$$

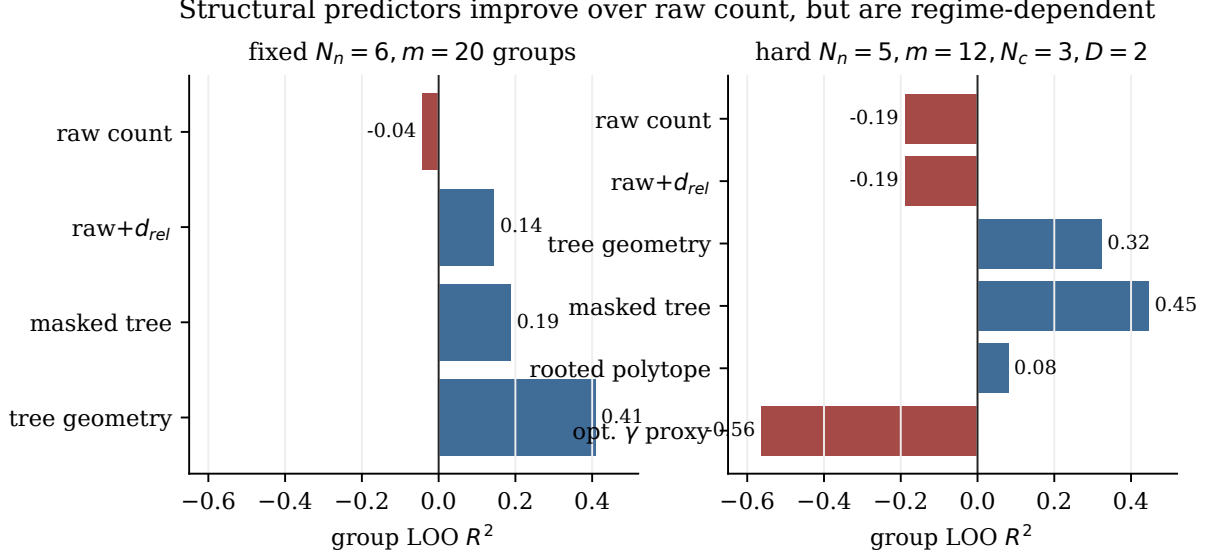


Figure 2: Structural predictors in two fixed-count regimes. Raw count is uninformative once fixed. Tree geometry and masked tree geometry explain residual group-level variation, but the best predictor depends on regime.

The input mask fixes which coordinates can enter which edge vectors:

$$\Omega_{e\alpha} = 0 \quad \Rightarrow \quad K_{e\alpha} = 0.$$

The real capacity object is therefore $\gamma^*(G, \Omega)$, not only $\gamma^*(G)$.

The mechanism-isolation file shows fixed-graph, fixed-count, fixed- d_{rel} cases where masks still change ICL. On the fixed random physical graph, changing edge-load structure gave +6.96 mean ICL points, while changing coordinate-load heterogeneity gave -16.48 mean ICL points. On the fixed cycle graph, the corresponding deltas were -6.32 and +2.12 points; on the fixed hub graph they were +3.44 and -1.52 points. The sign is not universal, but the effect exists.

[7] Post-training mechanism evidence

Pre-training topology asks what the graph could use. Post-training diagnostics ask what the trained model did use. The strongest diagnostic class was projection/root/tree organization around comparison branches. Projection-alignment summaries gave

$$\begin{aligned} R_{LOO}^2 &= 0.740 && \text{at run level,} \\ R_{LOO}^2 &= 0.809 && \text{at topology-mean level,} \\ R_{LOO}^2 &= 0.599 && \text{for topology-best seed.} \end{aligned}$$

This means high-ICL models are not merely high-count systems. They organize the available tree-geometry so that comparison branches produce decodable steady states. The active-tree and

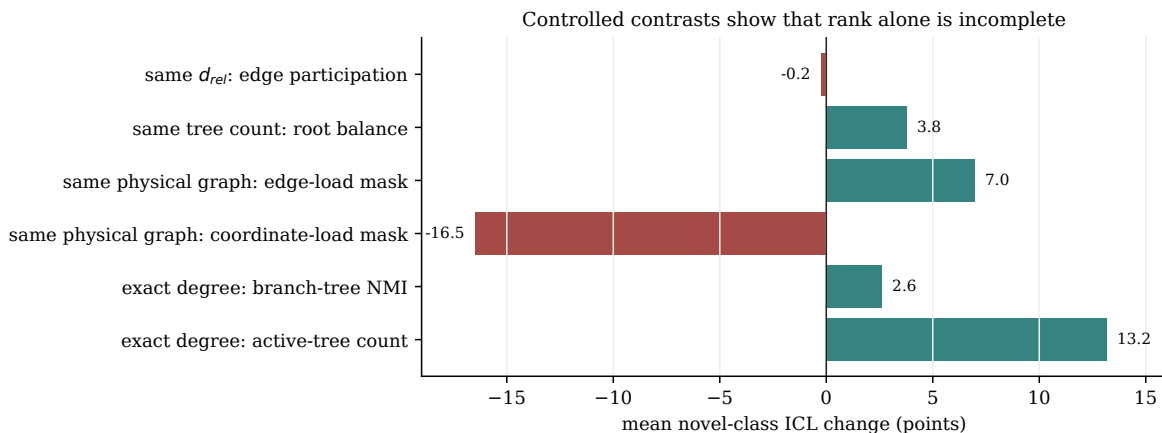


Figure 3: Controlled contrasts. Some coarse structural metrics are not monotone drivers, and fixed-graph input masks can still change ICL. This is why d_{rel} alone is incomplete.

Post-training projection organization is the strongest diagnostic

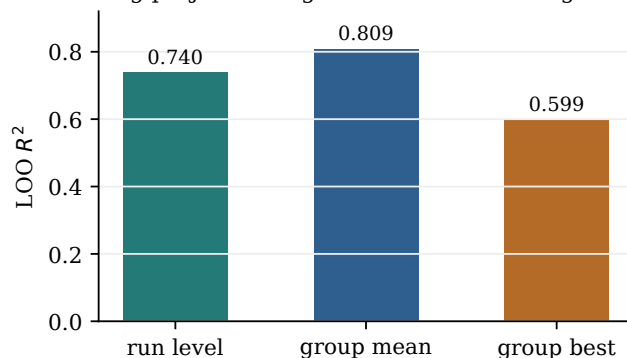


Figure 4: Post-training projection alignment is much more explanatory than coarse pre-training count. This is mechanism evidence, not an independent capacity theorem.

margin diagnostics are close to the learned mechanism, so they should be read as mechanistic explanations rather than clean pre-training predictors.

[8] Causal scrambles

The causal interventions are the strongest evidence that the learned alignment matters. Two statistic-preserving scrambles were central:

branch-alignment scramble	preserves graph, root tree counts, d_{rel} , and input density
projection scramble	also preserves input support and per-edge projection row norms

They break branch/projection/tree organization while preserving many coarse statistics.

In the stratified hard-regime pilot, high-ICL runs dropped by about -96 points under branch-alignment scrambling and about -64 to -68 points under projection scrambling. Low and mid

runs also collapsed substantially.

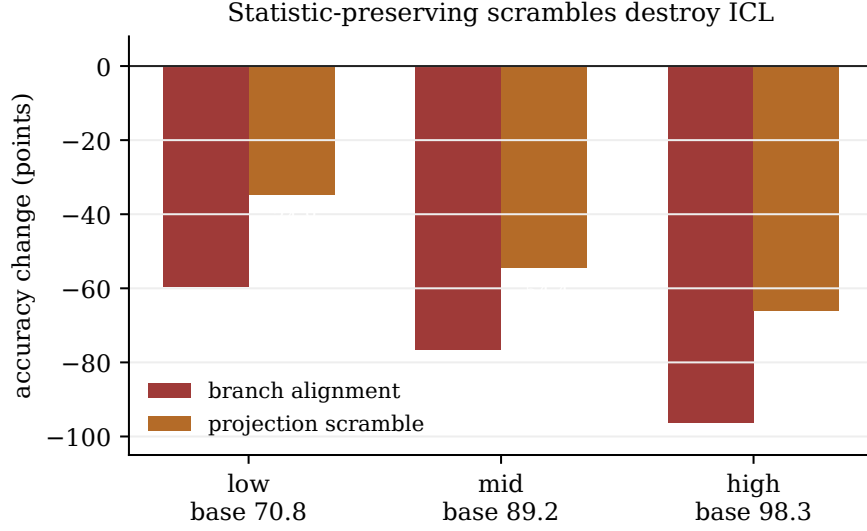


Figure 5: Statistic-preserving scrambles in low, mid, and high baseline buckets. Destroying branch/projection/tree alignment sharply reduces novel-class ICL.

This pushes the evidence beyond correlation. It says the trained CRNs rely on the organized relation between input branches, learned edge projections, and root/tree structure.

[9] Hard regimes and the normal-fan pilot

Three hard regimes were added: $n4_m6_N3_D2$, $n5_m8_N3_D2$, and $n5_m12_N3_D2$, each with 60 trained runs plus mechanism and causal follow-ups. The results are mixed in the useful sense. Topology signal depends on regime.

In $n5_m12_N3_D2$, masked tree geometry had LOO $R^2 = 0.447$ and positive held-out-family performance. In the sparser $n5_m8_N3_D2$ regime, within-sample bootstrap improvements were more positive than held-out-family prediction. This suggests that topology may have sparse, intermediate, and redundant regimes rather than one monotone scalar rule.

The exact-degree normal-fan pilot fixed

$$N_n = 5, \quad m = 12, \quad N_c = 3, \quad D = 2, \quad \text{in/out degree sequence} = [3, 2, 2, 3, 2], \quad d_{\text{rel}} = 88.$$

It varied tree-polytope/normal-fan diagnostics through directed degree-preserving rewires. Only four topology groups were trained, so this is not yet a statistical result. It is a constructive design showing that topology can vary beyond degree sequence and d_{rel} .

The descriptive $n = 4$ correlations were $r = 0.869$ for active-tree count and $r = 0.691$ for branch-tree NMI with mean ICL. These numbers are promising design feedback, not proof.

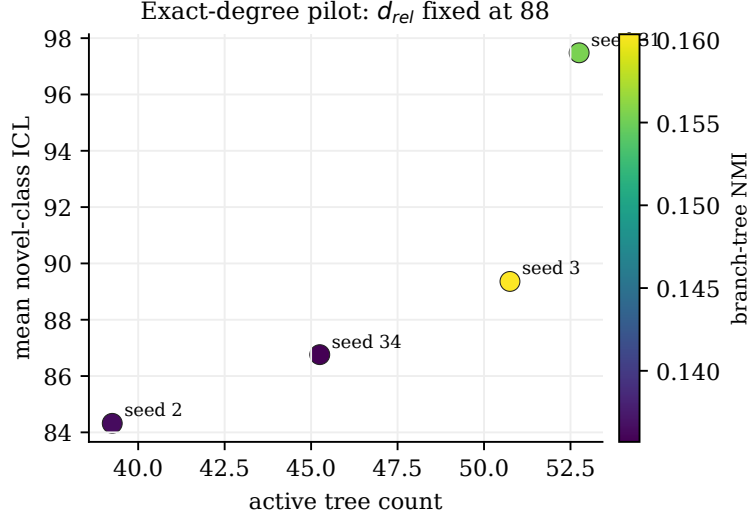


Figure 6: Exact-degree pilot. All points have the same degree sequence and $d_{\text{rel}} = 88$. Active tree count and branch-tree NMI vary across rewires, and mean ICL varies with them in this small $n = 4$ pilot.

[10] Motifs and matched controls

Essential physical subgraphs retrained better than sparse input masks under the extraction protocol. But matched controls softened the stronger interpretation. Degree-rewired and random strongly connected controls often matched or exceeded the extracted motifs.

The corrected interpretation is:

small physical subgraphs can support ICL, but the extracted motifs are not uniquely optimal.

This is compatible with the tree-sum view: many different graphs can provide usable branch-covering tree geometry.

[11] The gamma-star attempt

The project implemented γ^* -style capacity probes because d_{rel} is too coarse. This was the right theoretical direction, but the first optimized proxy did not outperform tree geometry. In the hard $n5_m12_N3_D2$ regression, optimized γ capacity had LOO $R^2 = -0.564$, while tree geometry had 0.324 and masked tree geometry had 0.447.

The next audit repaired the analytic toy gate. The no-bias probe now reproduces the paper’s small cases: two species with both branches fails, two species with the max branch passes, and the complete directed three-species both-branch construction passes. That repair validates the lower-tail object as a diagnostic. It does not make γ_{ICL}^* a predictor: in the fixed m20 reanalysis, tree-difference multiplicity reached mean-ICL LOO $R^2 = 0.435$, while the best repaired gamma model reached only 0.078.

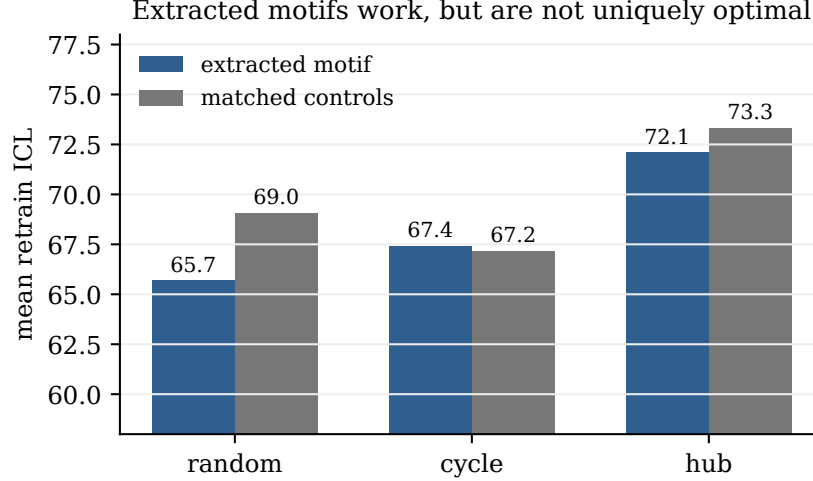


Figure 7: Extracted physical motifs support ICL, but matched controls are competitive. The evidence supports a family of feasible physical substrates, not one unique motif.

The lesson is sharper than simple failure. ICL is a worst-branch problem:

$$\max_{K,B} \min_{b \in \mathcal{B}} \text{margin}_b(K, B; G, \Omega).$$

A good capacity theory should target low-percentile or worst-branch separation under norm constraints, but the current repaired γ diagnostic still misses important topology/mask variables in trained regimes.

[12] The Markov-ICL extension

The next question was not “run a bigger sweep.” It was:

Can the first-order topology story be sharpened using Markov-process expressivity theory?

The requested ingredients were input multiplicity, monotonicity limits, coefficient constraints, non-equilibrium driving, branch sharpness, structural compatibility, and a clean separation between expressivity and trainability.

The exact representation from Section 2 is already the bridge. For first-order CRNs,

$$\tau_r(z) = \sum_{T \in \mathcal{T}_r(G)} \exp(\beta_T + \Theta_T^\top z), \quad \Theta_T = \sum_{e \in T} K_e.$$

Thus the Markov-expressivity objects must be defined on tree-sum features, not isolated edges.

The input-encoding multiplicity is

$$M_\alpha = \sum_e \Omega_{e\alpha}.$$

For ICL, the important refinements are branch-specific:

$$M_{i,d}, \quad M_{q,d}, \quad \sum_e \Omega_{e,i,d} \Omega_{e,q,d}, \quad |M_{i,d} - M_{q,d}|.$$

Low or imbalanced comparison-coordinate multiplicity is the monotonicity-risk diagnostic:

$$\text{monoRisk}_\alpha = \mathbf{1}[M_\alpha \leq 1].$$

After training, an effective participation diagnostic is

$$M_\alpha^{\text{eff}} = \frac{(\sum_e |K_{e\alpha}|)^2}{\sum_e K_{e\alpha}^2}.$$

Branch sharpness is distinct from branch coverage. For branch direction u_b ,

$$R_{r,b} = \max_{T \in \mathcal{T}_r(G)} \Theta_T^\top u_b - \min_{T \in \mathcal{T}_r(G)} \Theta_T^\top u_b.$$

This measures whether a rooted tree polytope can create a sharp margin along a comparison direction. A graph can cover every branch and still have weak lower-tail margins.

Coefficient controllability is the other missing piece. Tree-polynomial coefficients are constrained by overlapping spanning trees; they are not freely adjustable. Therefore rank, even masked rank, can overestimate useful capacity. Effective rank, condition number, tree entropy, normal-fan coverage, extremal-tree accessibility, and branch-specific concentration are all relevant.

The theory note produced in this phase is `ICL/markov_icl_expressivity_theory.md`. Its shortest message is:

First-order Markov-ICL expressivity is tree-sum expressivity under an input mask. Input multiplicity and branch sharpness are useful structural variables, but not standalone laws.

[13] Existing-data Markov reanalysis

The reanalysis script `ICL/markov_expressivity_reanalysis.py` consumes committed CSV/JSON artifacts only. It does not launch new broad training. It groups seeds by topology/mask before inference.

Coverage:

dataset	rows	topology/mask groups
fixed m20	240	48
hard $n4_m6$	60	12
hard $n5_m8$	60	12
hard $n5_m12$	60	12
exact-degree pilot	4	4
exact-degree library	32	32

In the fixed m20 data, grouped LOO R^2 for mean novel-class ICL was 0.095 for aggregate multiplicity variables, 0.046 for comparison-multiplicity variables, 0.158 for tree-geometry variables, and 0.145 for the old capacity proxy. In the hard $n5_m12$ data, tree geometry reached 0.437 while aggregate multiplicity and the old capacity proxy were -0.190 .

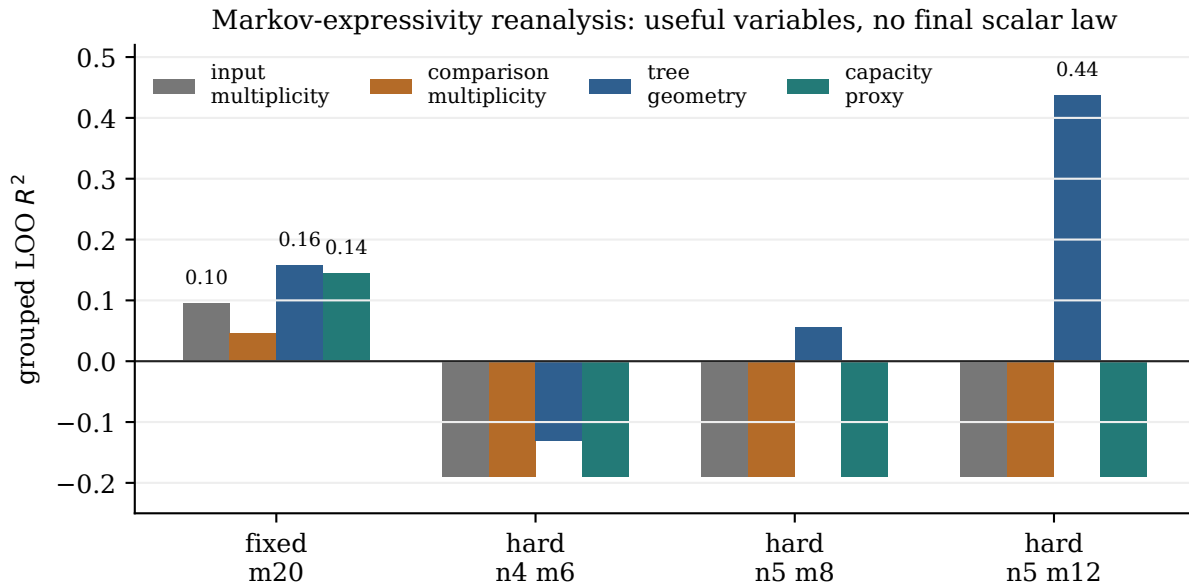


Figure 8: Existing-data Markov reanalysis. The richer Markov variables are useful diagnostics, but the evidence still does not support a universal scalar expressivity law.

The input-multiplicity control is especially clarifying. In the fixed-count mask data, all mask families have mean aggregate multiplicity $M_{\text{mean}} = 10$. Yet the coordinate-block family has zero minimum comparison overlap, high coordinate-load Gini, and much lower mean ICL: 63.4 points versus roughly 74–77 points for the other families.

The exact-degree normal-fan pilot was also reinterpreted through this lens. It has only four trained groups, but those groups share the intended fixed-degree and fixed- d_{rel} controls. Descriptive correlations with mean ICL were $r = 0.869$ for normal-fan active tree count, $r = 0.691$ for branch-tree NMI, $r = 0.810$ for log tree count, and $r = 0.759$ for effective rank. These are design signals, not statistical proof.

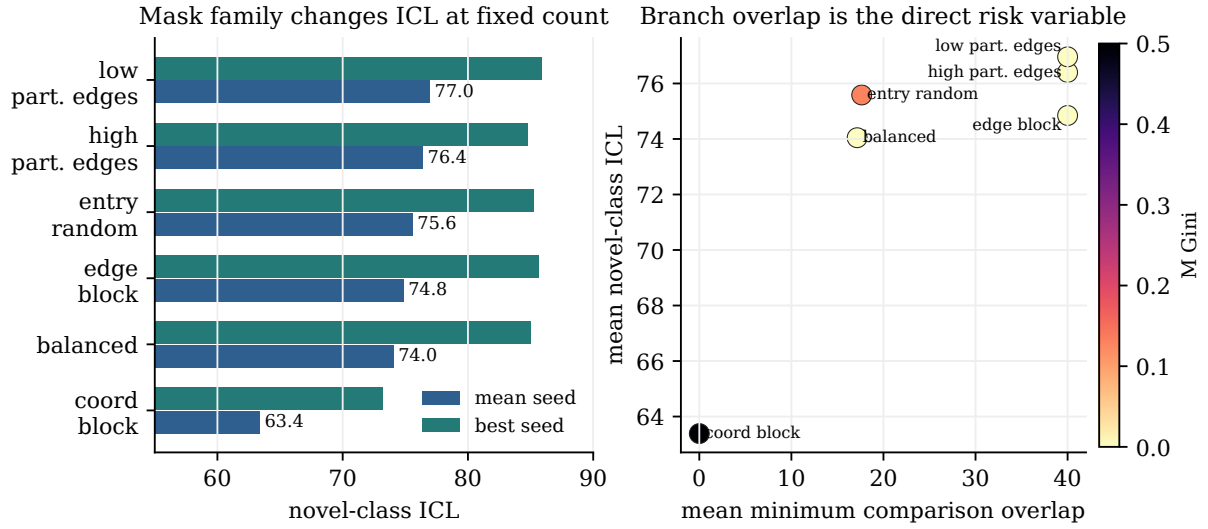


Figure 9: Input multiplicity control from existing data. Average multiplicity alone is not the law; branch comparison overlap and load imbalance are the direct failure risks.

[14] Improved lower-tail capacity

The improved probe is `ICL/branch_margin_capacity_v2.py`. It targets the lower tail directly:

$$\gamma_{\text{ICL}}^*(G, \Omega; \mathcal{C}) = \max_{\theta} \min_{b \in \mathcal{B}} \text{LCVaR}_{\alpha} [m_{\theta}(z) \mid z \in b],$$

where

$$m_{\theta}(z) = q_{y(z)}(z) - \max_{\ell \neq y(z)} q_{\ell}(z).$$

The implemented variants are exact log-sum-exp tree features, tropical max-over-tree features, and hard-root structural compatibility. The constraints include row-norm bounds for K_e , decoder Frobenius norm bounds, edge-bias bounds, and the input mask Ω .

A small smoke run on a 5-node, 12-edge cycle-chords graph produced finite branch-wise reports for all three variants. The objectives were still negative: -3.001 for exact, -2.800 for tropical, and -2.270 for hard-root. This validated the reporting path and reinforced that average accuracy is not the right target; the worst branch can still fail.

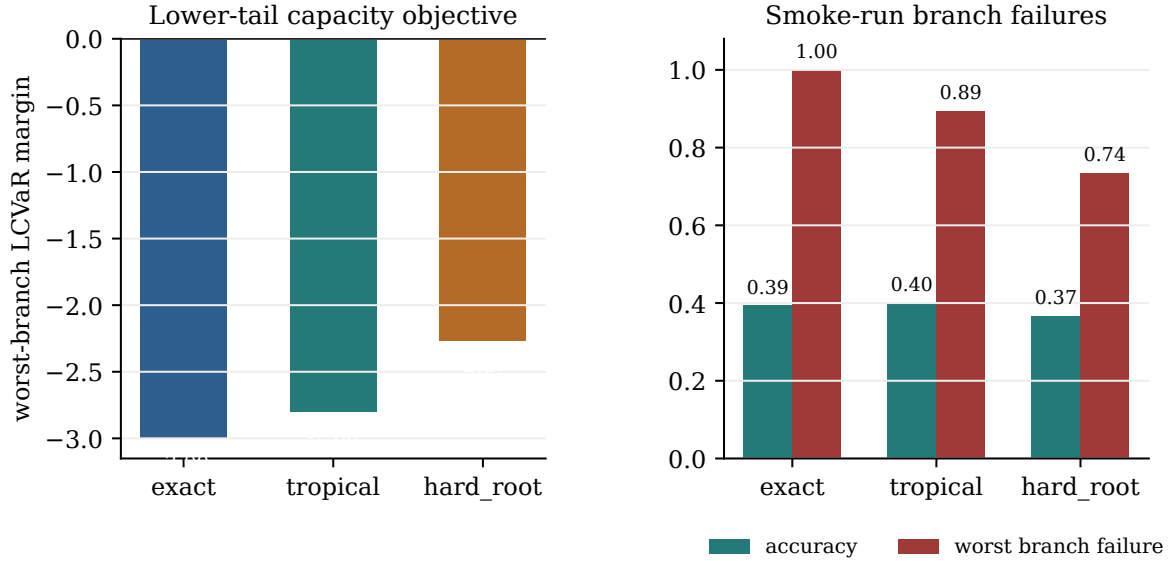


Figure 10: Lower-tail capacity V2 smoke result. The new probe reports the correct object: worst-branch lower-tail margin and branch failures. This smoke run is not an optimized capacity theorem.

The later toy repair passed the analytic gate, but the existing-data reanalysis remained negative for prediction. Repaired no-bias exact, tropical, and hard-root gamma variants had fixed-m20 mean-ICL LOO $R^2 = -0.075, -0.115, \text{ and } -0.150$, respectively. Combining repaired gamma with tree-difference multiplicity degraded the fixed-m20 mean-ICL model from 0.435 to 0.078. Thus γ_{ICL}^* should remain a diagnostic until it predicts held-out exact-control outcomes.

[15] Expressivity, trainability, and thermodynamics

The reanalysis keeps three outcomes separate:

$$\begin{aligned} \text{best seed} &\approx \text{expressivity envelope}, \\ \text{mean seed} &\approx \text{trainability/reliability}, \\ \text{seed variance} &\approx \text{optimization instability}. \end{aligned}$$

Across the 36 hard-regime groups, masked effective rank correlated 0.895 with best seed and 0.897 with mean seed, but only 0.345 with seed standard deviation. Trained margin and tree entropy were also strongly correlated with best/mean seed, but they are post-training mechanism descriptors rather than pre-training capacity metrics.

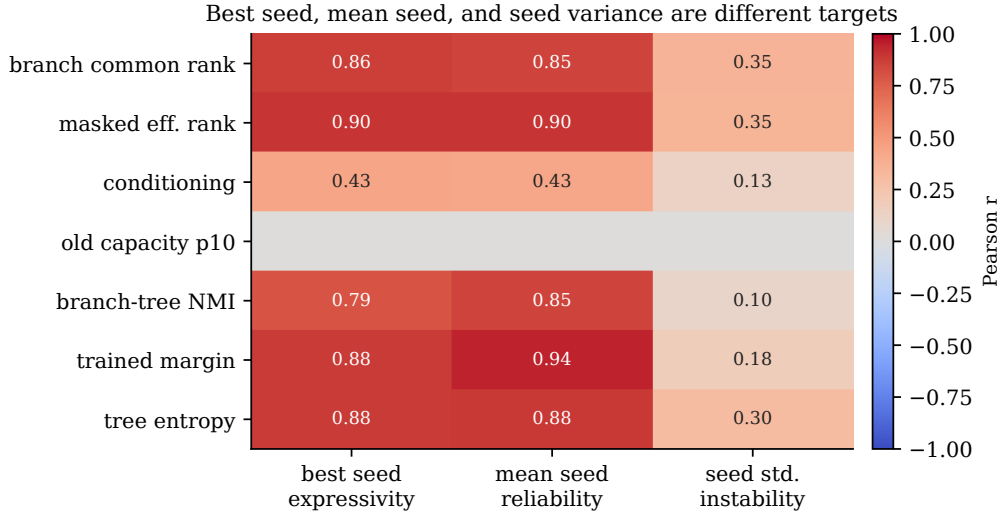


Figure 11: Expressivity and trainability should not be collapsed into one outcome. Best seed, mean seed, and seed standard deviation respond differently to structural and trained-mechanism variables.

The thermodynamic question is deliberately unresolved. Existing arbitrary directed exponential-rate models may be non-equilibrium, but they were not parameterized as reversible-edge thermodynamic Markov processes. Therefore the report `thermodynamic_force_budget_report.md/json` does not claim entropy production or a force-budget law. It only audits eligibility: among 36 local hard topology groups with topology JSON, the mean reversible-edge fraction is 0.512, with range 0 to 1.

A valid thermodynamic experiment needs a reversible support and a parameterization such as

$$W_{ij} = \exp(E_j - B_{ij} + F_{ij}/2 + \text{input drive}), \quad B_{ij} = B_{ji}, \quad F_{ij} = -F_{ji}, \quad |F_{ij}| \leq F_{\max}.$$

Only then should we sweep $F_{\max}$ and ask whether ICL branch margins increase with allowed non-equilibrium driving.

[16] Post-gamma exact-control phase

The next phase was deliberately gated. It did not launch another broad topology sweep. It first reconciled predictor names, lifted input multiplicity into tree and tree-difference space, repaired the analytic γ_{ICL}^* toy gate, and only then ran prospective exact controls.

The fixed-m20 tree-level multiplicity reanalysis was strong:

outcome	edge multiplicity	tree multiplicity	tree-difference multiplicity
mean novel-class ICL	-0.002	0.403	0.435
best-seed novel-class ICL	0.109	0.245	0.419

This supports the statement that useful input multiplicity lives in rooted-tree and tree-difference space, not merely edge space.

The repaired gamma gate passed the analytic small cases, but the existing-data prediction test was negative. In fixed m20, tree-difference multiplicity reached mean-ICL LOO $R^2 = 0.435$, while the best repaired gamma model reached only 0.078. Repaired exact, tropical, and hard-root gamma alone were all negative.

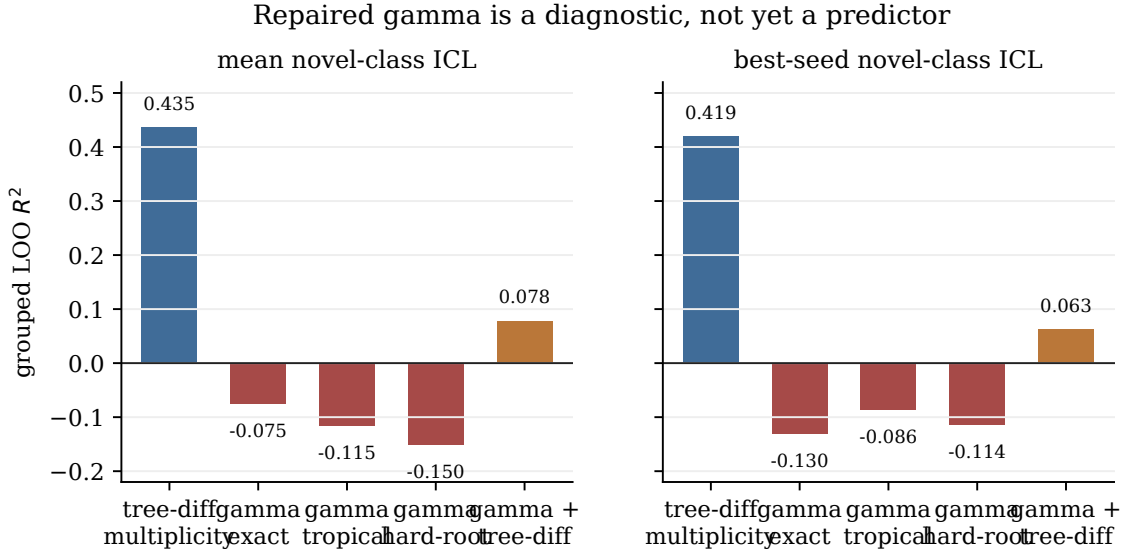


Figure 12: Repaired no-bias gamma passes analytic toys but does not yet predict trained ICL in existing data. Tree-difference multiplicity remains stronger in fixed m20.

The first prospective tree-difference causal control then fixed one physical graph, input-coupled count, d_{rel} , aggregate multiplicity, edge load, and coordinate-load stratum. It trained 16 mask groups with five seeds each. The intended high-overlap contrast did not improve ICL:

stratum	outcome	high overlap	low overlap	high-low
balanced load	mean ICL	75.360	79.330	-3.970
balanced load	best ICL	84.900	89.700	-4.800
imbalanced load	mean ICL	68.830	69.900	-1.070
imbalanced load	best ICL	74.450	76.250	-1.800

The balanced mean-ICL CI was $[-7.9905, 0.6700]$, and the balanced best-ICL CI was $[-9.000, -0.150]$.

Prospective exact control did not support tree-difference as a standalone knob

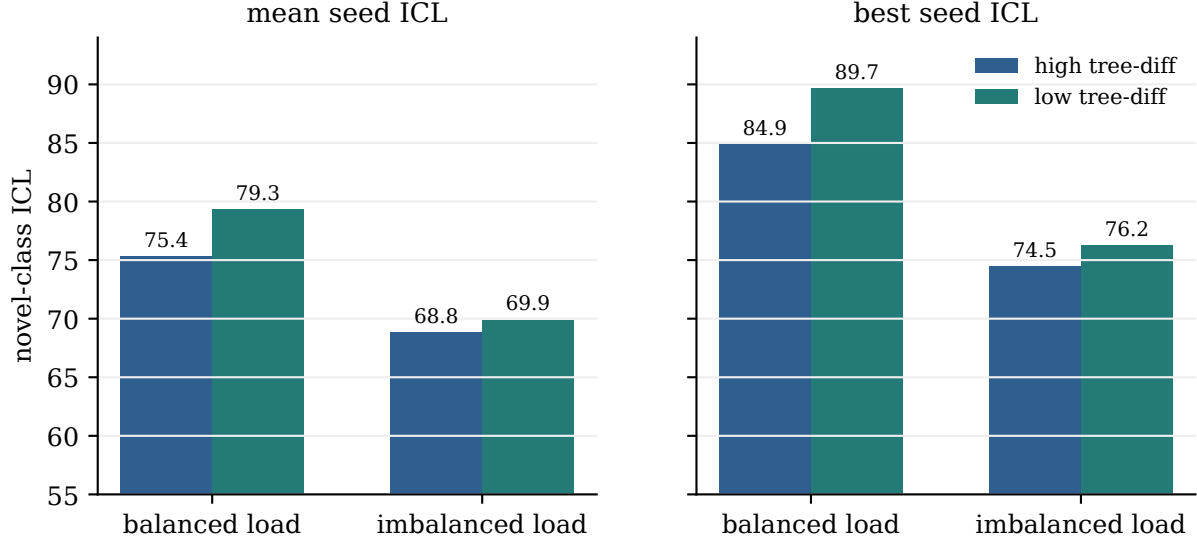


Figure 13: Prospective tree-difference exact control. High same-root tree-difference comparison overlap did not causally improve novel-class ICL under the first balanced-load exact-control design.

Grouped LOO tells the same story. For mean ICL, controls alone reached 0.488, tree-difference plus controls reached 0.447, and gamma plus controls reached 0.320. For best ICL, controls alone reached 0.601, tree-level plus controls reached 0.631, tree-difference plus controls reached 0.545, and gamma plus controls reached 0.571.

[17] Exact-degree normal-fan expansion

The exact-degree normal-fan pilot was then scaled. The expanded experiment fixed

$$N_n = 5, \quad m = 12, \quad N_c = 3, \quad D = 2,$$

the exact in-degree and out-degree sequence, full input coupling, $d_{\text{rel}} = 88$, and the full-coupling multiplicity distribution. It trained 32 topology groups with five seeds per group, for 160 runs.

This is now the best pre-training signal, but it is weak. Grouped LOO values were:

model	mean ICL	best ICL	seed std.
active-tree count	0.099	0.091	-0.069
normal-fan pair	0.112	0.110	-0.101
tree count	0.114	0.093	-0.069
gamma exact	-0.127	-0.072	-0.105
gamma + normal fan	0.059	0.117	-0.126

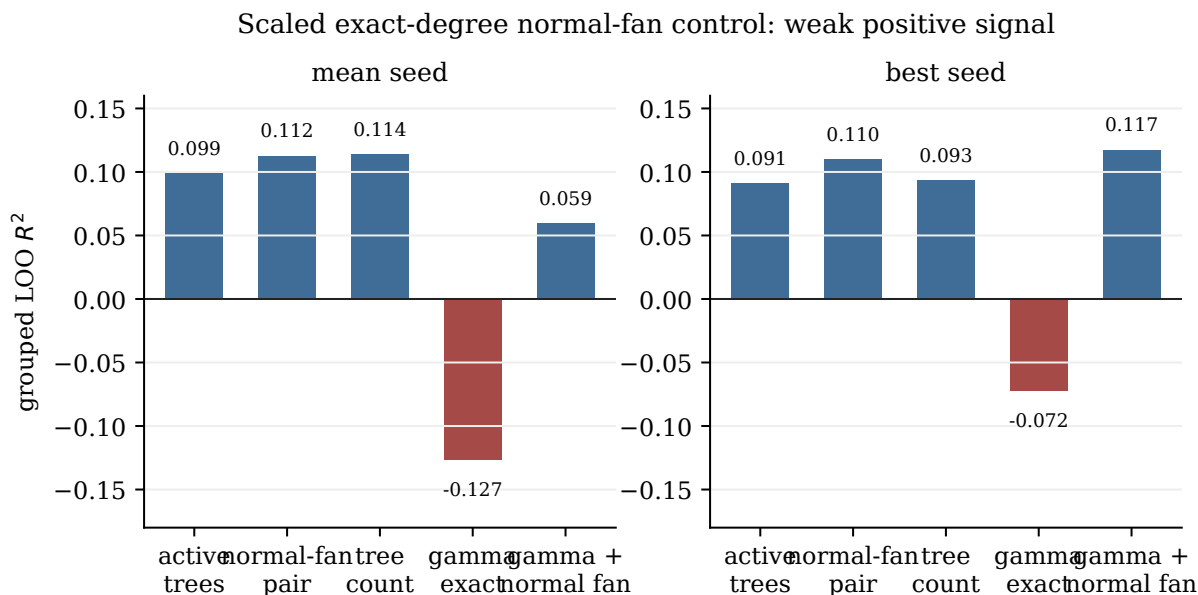


Figure 14: Scaled exact-degree normal-fan experiment. Active-tree/tree-count variables are the best current pre-training signal under exact controls, but the effect is small and does not justify a universal scalar law.

Correlations were clearer than LOO prediction: active-tree count correlated about 0.446 with mean ICL and 0.441 with best ICL; log rooted-tree count correlated about 0.460 with mean ICL and 0.439 with best ICL. This suggests that the next experiment must separate total rooted-tree count from genuine normal-fan branch geometry across multiple base graphs or degree sequences.

[18] Updated mechanism evidence

The exact-control phase also added post-training mechanism and causal intervention checks for the prospective-control models. Mechanism metrics were collected for all 80 runs. Four selected high-performing models were then subjected to statistic-preserving scrambles and selected edge ablations.

The causal drops were large:

intervention	mean accuracy change
context-block shuffle	−71.458
stat-preserving projection scramble	−59.688
stat-preserving branch-alignment scramble	−57.778
decoder-root permutation	−53.125

Selected edge ablations also showed dependence on input-coupled and physical edges: mean max input-ablation loss was 21.146 points, and mean max physical-edge ablation loss was 26.563 points.

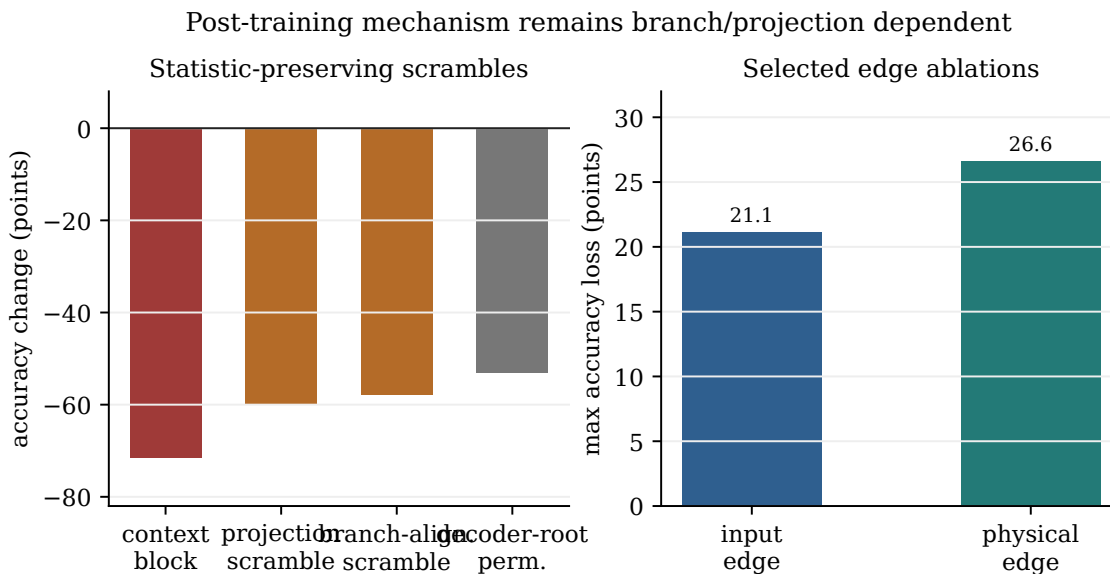


Figure 15: Mechanism follow-up after the prospective exact-control phase. Trained successful models still depend on branch/projection organization and ablation-sensitive edges.

This is an important separation. The prospective pre-training tree-difference overlap contrast was negative or inconclusive, but trained models that do work still causally depend on branch/projection organization. The mechanism claim is stronger than the current pre-training selector claim.

[19] Pre-multibase synthesis and motivation

The current claim separation is:

category	current status
Expressivity	First-order tree-sum basis is exact. Normal-fan active-tree/tree-count variables weakly predict best-seed ICL under exact controls.
Trainability	Mean-seed ICL follows the same weak normal-fan/tree-count signal, but seed variance is not explained well.
Mechanism	Trained models rely on branch/projection/tree organization; statistic-preserving scrambles collapse selected trained models.
Causal evidence	The first prospective tree-difference causal control did not support tree-difference overlap as a standalone causal knob.
Thermodynamics	Untested. No $F_{\max}$ claim should be made until a reversible-support parameterization is implemented.

Before the completed multi-base analysis, the concrete next task was no longer to prove that topology matters broadly. The task was:

diagnose why the fixed-m20 tree-difference signal failed under prospective exact control, and test whether normal-fan active-tree/tree-count geometry survives multi-base exact controls after separating it from total rooted-tree count.

The completed multi-base phase therefore did or preserved the following:

1. treat same-root tree-difference overlap as a failed standalone causal knob, not as the main selector;
2. use cross-root and decoder-aware contrast metrics as diagnostics, not as proof;
3. consume the multi-base exact-degree library that separates rooted-tree count from normal-fan geometry;
4. keep repaired γ_{ICL}^* as a diagnostic rather than a selector;
5. keep thermodynamics delayed until reversible support and detailed-balance behavior at $F_{\max} = 0$ are verified.

The defensible pre-multibase statement was:

In tested first-order regimes, topology and masks shape the rooted tree-sum basis used for ICL. Existing-data tree-difference multiplicity is predictive, but the first prospective exact-control contrast did not support it as a standalone causal knob. The best pre-multibase signal was weak normal-fan/tree-count geometry, while the strongest mechanism evidence remained post-training branch/projection dependence.

[20] Completed multi-base exact-control analysis

The May 13 handoff library has now been consumed and analyzed. The file

`ICL/multibase_normal_fan_tree_count_library.py`

built a multi-base exact-degree library whose purpose was to separate total rooted-tree abundance from task-aligned normal-fan/active-tree geometry. This remained a targeted exact-control design, not a broad sweep.

The controls were

$$N_n = 5, \quad m = 12, \quad N_c = 3, \quad D = 2, \quad p = 8, \quad d_{\text{rel}} = 88,$$

with full input coupling, exact multiplicity $M_\alpha = m$ for every input coordinate, and exact in/out degree sequence within each base graph. The design used seven base graph families, 80 directed degree-preserving rewires per base, and paired selection arms:

arm	pairs	mean $ \Delta \text{ tree count} $	mean $ \Delta \text{ normal-fan score} $
fixed tree count, variable normal fan	11	0.909	4.353
variable tree count, matched normal fan	9	14.667	0.060

The Shurm result audit is complete: 185 of 185 training result files exist, covering all 37 selected topology groups with five seeds each. The analysis artifacts are recorded in

`ICL/results/next_phase_stats/multibase_goal_completion_audit.md/json.`

The local result table has 37 topology-group rows; seed rows were aggregated before inference.

The grouped LOO result is stronger than the earlier one-base exact-degree experiment. For mean novel-class ICL and best-seed novel-class ICL:

model	mean ICL R^2_{LOO}	best ICL R^2_{LOO}
base controls only	0.431	0.793
tree count + base	0.561	0.834
normal fan + base	0.518	0.796
tree count + normal fan + base	0.499	0.788
cross-root + tree count + normal fan + base	0.446	0.787
gamma exact + base	0.375	0.797
gamma + tree count + normal fan + base	0.470	0.806

The best grouped-LOO model in this analysis is therefore **tree_count_plus_base**: $R^2 = 0.561$ for mean ICL and $R^2 = 0.834$ for best-seed ICL. This is the clearest current pre-training signal. It should be read as rooted-tree abundance plus base-graph controls, not as a universal graph law.

The paired arms give the same qualitative update. Arm A, which changes normal-fan geometry at nearly fixed tree count, was weak:

$$\Delta_{\text{high-low}} \text{ mean ICL} = +1.193, \quad 95\% \text{ CI} = [-1.291, 4.007].$$

Next exact-control library separates rooted-tree abundance from normal-fan geometry

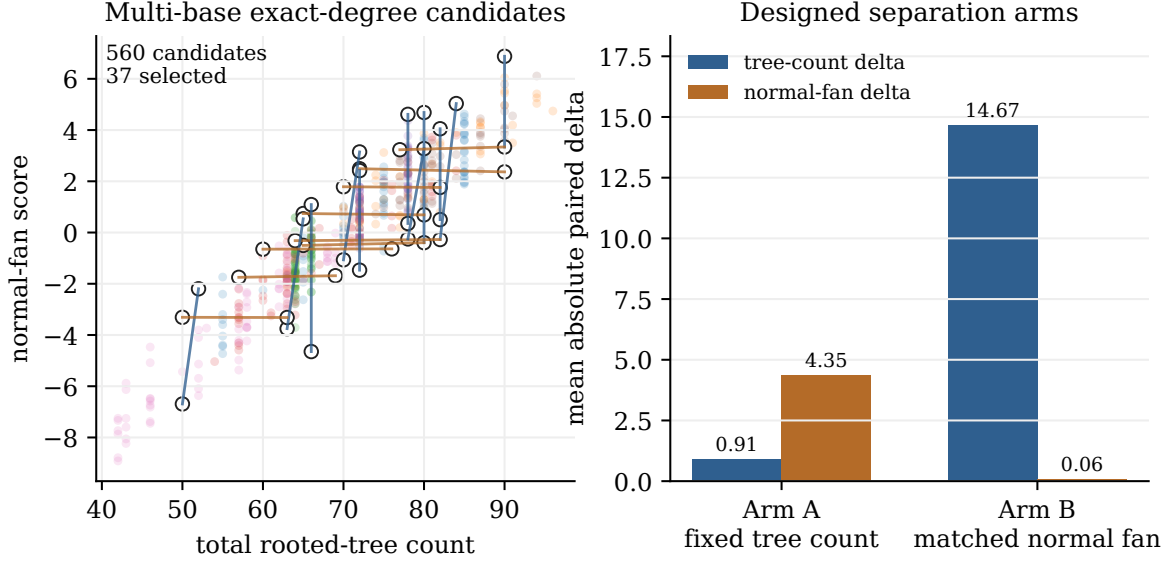


Figure 16: Multi-base exact-control design. Arm A holds rooted-tree count nearly fixed while varying normal-fan score. Arm B varies rooted-tree count while matching normal-fan score.

Arm B, which changes rooted-tree count at matched normal-fan score, was stronger:

$$\Delta_{\text{high-low}} \text{ mean ICL} = +4.067, \quad 95\% \text{ CI} = [0.876, 7.364].$$

Best-seed paired contrasts were smaller: +0.764 for Arm A and +0.556 for Arm B, with intervals crossing zero.

There is an important caveat. Held-out-base prediction is not yet positive: the held-out-base R^2 values are -0.157 for mean ICL, -0.136 for best ICL, and -0.410 for seed standard deviation. Within-base residualized correlations are more favorable: for mean ICL, log rooted-tree count has $r = 0.545$, active-tree count has $r = 0.500$, and cross-root effective-rank summaries have $r = 0.531$. This means the current evidence supports a within-exact-control topology signal, especially rooted-tree abundance, but not a base-universal capacity law.

[21] Cross-root diagnostics, gamma, and multi-base mechanisms

The same-root tree-difference failure motivated cross-root and decoder-aware diagnostics. The reason is structural: the steady state normalizes all rooted-tree numerators,

$$\bar{C}_r(z) = \frac{\tau_r(z)}{\sum_s \tau_s(z)},$$

and the decoder compares species/root concentrations. The implemented file

ICL/cross_root_decoder_contrast_metrics.py

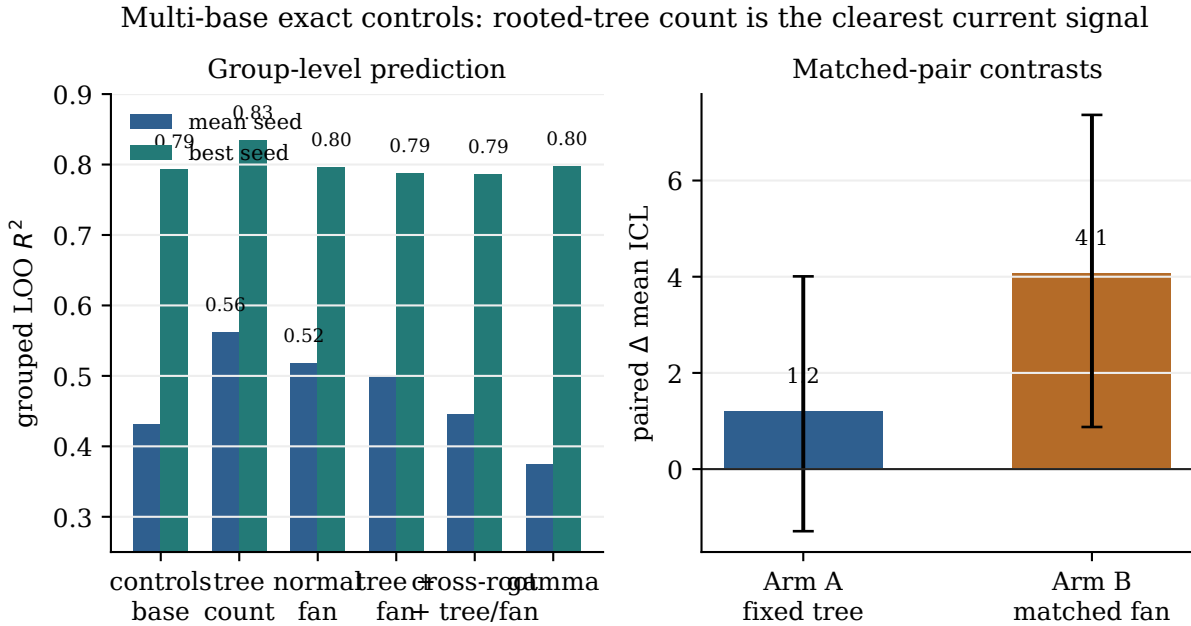


Figure 17: Completed multi-base exact-control analysis. Tree count plus base controls gives the best grouped LOO prediction. The matched-pair contrast is clearer for Arm B, where tree count varies at matched normal-fan score.

therefore adds cross-root tree-difference participation and decoder-agnostic root-pair coverage summaries.

In the completed multi-base library, full input coupling makes binary cross-root overlap essentially saturated. The varying cross-root quantities are rank/effective-rank and root-pair diversity proxies. Their marginal correlations with ICL were high in the pooled table, about $r = 0.744$ with mean ICL and $r = 0.745$ with best ICL for the cross-root effective-rank summaries. However, adding cross-root summaries to the tree-count/normal-fan/base model reduced grouped LOO performance to 0.446 for mean ICL and 0.787 for best ICL. Cross-root metrics are therefore useful diagnostics, not yet better selectors.

Repaired no-bias gamma remains in the same category. In this multi-base analysis, `gamma_exact + base` had $R^2_{LOO} = 0.375$ for mean ICL and 0.797 for best ICL. Adding gamma to tree count and normal fan gave 0.470 for mean ICL and 0.806 for best ICL, still below tree count plus base alone. Gamma is analytic-toy valid and scientifically useful, but it still should not be used as a standalone topology selector.

The strongest evidence channel remains post-training mechanism. Eight selected multi-base trained models were subjected to mechanism diagnostics and statistic-preserving causal scrambles, giving 64 scramble rows. The mean accuracy changes were:

intervention	mean accuracy change
context-block shuffle	−79.562
stat-preserving projection scramble	−55.031
stat-preserving branch-alignment scramble	−54.406
decoder-root permutation	−48.187

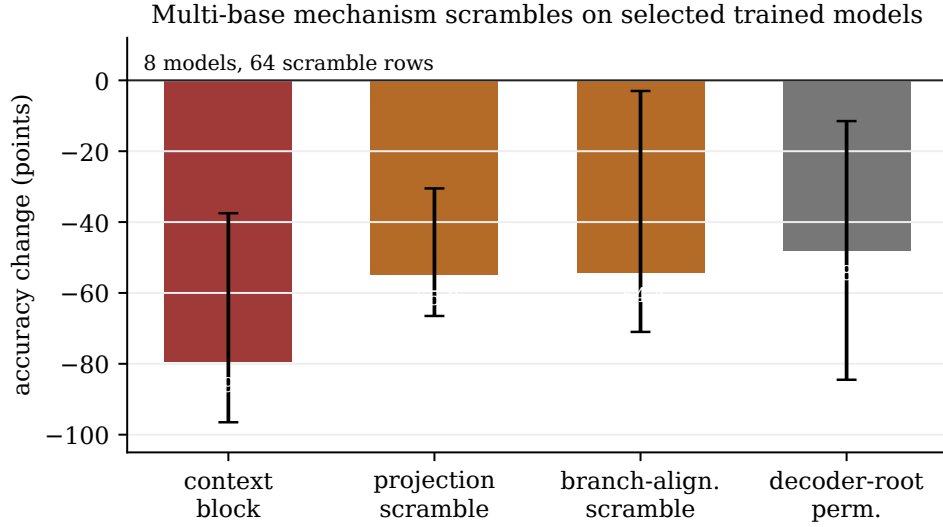


Figure 18: Multi-base mechanism follow-up. Even when pre-training predictors remain imperfect, trained high-performing models depend causally on branch/projection/tree organization.

The current claim separation is now:

category	current status
Exact theory	First-order CRNs compute through rooted tree-sum projections exactly by the matrix-tree theorem.
Pre-training prediction	Rooted-tree count plus base controls is the clearest current signal under multi-base exact controls; held-out-base performance is still negative.
Expressivity	Best-seed ICL is predicted best by tree count plus base in this library, but this is not yet a base-universal law.
Trainability	Seed standard deviation remains poorly predicted; the best reported seed-std models are negative in grouped LOO.
Mechanism	Trained models rely on branch/projection/tree organization; statistic-preserving scrambles collapse selected multi-base models.
Causal evidence	Same-root tree-difference overlap failed as a standalone prospective causal knob. Arm B now supports rooted-tree abundance as the better current exact-control signal.
Thermodynamics	Untested. No $F_{\max}$ or entropy-production claim is supported.

The most defensible updated statement is:

In tested first-order regimes, topology and masks shape the rooted tree-sum basis used for ICL. Same-root tree-difference multiplicity was predictive retrospectively but failed as a standalone prospective causal knob. In the completed multi-base exact-control experiment, rooted-tree abundance plus base controls is the best current pre-training signal, while post-training branch/projection/tree dependence remains the strongest evidence channel. No universal scalar topology law or thermodynamic force-budget law has been established.